1 Read: KBo 41.22, an orphan (CTH 470)
3′ i-da-a-lu-un UD-an ma-
4′ DINGIRMEŠ-aš kar-pí-in pa-an-ga-u-wa-aš EME-an
5′ ša-me-ed-du-wa-ra-at-kán ḫi-x
6′ na-aš-ta EN SÍSKUR ŠUM-ŠU
7′ le-e ku-wa-pí-ik-ki a-x
8′ te-eš-ḫa-an ma-ni-ku-wa-an(-)
9′ nu-uš-ši GIpát-tar pa-ra-a
10′ MUNUSŠU.GI-ma me-ma-i x
11′ DINGIRMEŠ-aš kar-pí-in pa-an-ga-u-wa-aš
12′ LÚA.ZU-ma-aš-ši-iš-ša-an
13′ na-aš ša-ra-a ti-
14′ na-aš-ta GU4-un
15′ a-ra-aḫ-za-an-da
function verb realia agent, deity place animal, body, nature abstract adjective foreign unclear
2 Count
Each word becomes several features, its form and its sign sequences:
pa-an-ga-u-wa-aš
→ word pankauaš 0.18
→ signs pa-an-ga 0.16
→ signs an-ga-u 0.16
→ signs ga-u-wa 0.17
A feature weighs more if it is frequent here and rare in the corpus:
$x_f \propto (1 + \log tf_f) \cdot idf_f$
idf of pankauaš 6.8; of the sign na 1.2
The fragment is a vector $\mathbf{x}$ with 177 of 25,372 features.
3 Compare
Each tradition is the average vector $\mathbf{c}_t$ of its manuscripts. Similarity is the cosine:
$s_t = \cos(\mathbf{x}, \mathbf{c}_t) = \sum_f x_f c_{t,f}$
Luwian rituals 0.389
Kizzuwatna rituals 0.245
Hattian cult layer 0.204
Syro-Mesopotamia 0.181
Kizzuwatna–Lower Land 0.173
Palaic festivals 0.165
Luwian festivals 0.165
Ḫuwaššanna festivals 0.160
Mixed Hurrian 0.147
Arzawa 0.146
4 Decide
The lead of the nearest tradition over the runner-up:
$m = s_1 - s_2 = 0.39 - 0.25 = 0.14$
Placed only if $m \geq \tau = 0.018$, the lead at which 80% of test fragments are placed correctly; otherwise left unassigned.
⇒ Luwian rituals
5 Explain
The lead splits exactly over the features:
$m = \sum_f x_f (c_{1,f} - c_{2,f})$
feature $x_f$ $c_{1,f}$ $c_{2,f}$ share ×10³
EMEan 0.11 0.17 0.00 +18.1
EME-an 0.10 0.17 0.00 +17.1
pankauaš 0.18 0.08 0.00 +14.7
ga-u-wa 0.17 0.08 0.00 +12.7
an-ga-u 0.16 0.07 0.00 +11.8
pa-an-ga 0.16 0.07 0.00 +11.8
EN 0.05 0.00 0.06 −3.0
EN 0.04 0.00 0.06 −2.3
$c_1$ Luwian rituals, $c_2$ Kizzuwatna rituals; bold: word form, italic: signs.
6 Check by hand
3′ i-da-a-lu-un UD-an ma-
4′ DINGIRMEŠ-aš kar-pí-in pa-an-ga-u-wa-aš EME-an
5′ ša-me-ed-du-wa-ra-at-kán ḫi-x
6′ na-aš-ta EN SÍSKUR ŠUM-ŠU
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Darker: larger share of the lead.
The formula pa-an-ga-u-wa-aš EME-an places it. Of the five nearest single manuscripts, 5 are Tunnawiya's ritual (CTH 409), which no reference group includes: a lead for the philologist.